

RISK-ADJUSTED PORTFOLIO BENCHMARKING

Quantitative review and rebuilt methodology

A critical rebuild of the latent-performance benchmark for the 25 Fama-French size and book-to-market portfolios.

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Review date: 16 July 2026

Sample: July 1926 to November 2025

Status: validated research implementation

This report supersedes the pre-rebuild document. The invalid old report and outputs have been removed and must not be cited.

1 Executive conclusion

The rebuild changes both the method and the strength of the claims. The primary score is now Fama-French three-factor alpha with 12-lag Newey-West inference and cross-sectional empirical-Bayes shrinkage. Stochastic frontier analysis (SFA) is retained only as a secondary test of one-sided residual asymmetry.

The validated panel has 29,825 unique observations: 25 portfolios by 1,193 months. No duplicates, missing calendar months, factor disagreement, or panel imbalance remain.

- SMALL HiBM ranks first at 117 posterior bps/year, but its interval includes zero. Only BIG LoBM has a posterior interval wholly above zero.
- Point ranks are much more precise-looking than the evidence. Common-date block-bootstrap rank intervals are wide for most portfolios.
- Forward validation produces a modest mean rank correlation of 0.096 and a top-minus-bottom future alpha spread of 172 bps/year.
- Only 2 of 25 portfolios support a one-sided SFA component at 5%; unsupported SFA ranks are suppressed.
- Residual diagnostics reject normality for 25/25 portfolios, lag-12 independence for 16/25, and no ARCH effects for 25/25.

Decision implication: use the posterior scores as uncertain historical benchmark diagnostics. Do not interpret the ranking as a stable skill hierarchy, and do not use SFA scores as a general efficiency league table.

2 Data integrity and review corrections

The pre-rebuild merged file contained 243,950 rows but only 29,825 unique portfolio-month keys. The implied inflation factor was 8.18, and downstream tables reported 9,758 observations per portfolio where only 1,193 months existed. That was a data error, not additional information.

Check	Validated result
Unique portfolio-months	29,825
Duplicate keys	0
Portfolios x months	25 x 1,193
Balanced panel	True
Missing calendar months	0
Sample	July 1926 - November 2025

The loader now fails on duplicate keys, null required values, factor disagreement within a month, an unbalanced panel, or a gap in the calendar sequence. Every run records SHA-256 hashes and byte sizes for both source files.

Portfolio file SHA-256: 05b45903b4ece361e80733136fc6e13a6e475a9c18b096d86345be458f210061

Factor file SHA-256: 3a412f4bb89d740af47ba331ca9f9b7d9fa0657cf22fa3c5f142d0db8ce68c70

Source: Kenneth R. French Data Library, with the official 25 size/book-to-market portfolio construction documentation. The local review files are treated as immutable inputs and identified by hash.

3 Primary estimand and uncertainty

For portfolio i and month t , the benchmark regression is:

$$r_{it} - r_{ft} = \alpha_i + \beta_i' f_t + \epsilon_{it}$$

The factors are market excess return, SMB, and HML. A 12-lag Bartlett-kernel Newey-West covariance matrix protects coefficient inference against heteroskedasticity and autocorrelation. Raw alpha p-values are adjusted across the 25 portfolios with the Benjamini-Hochberg false-discovery procedure.

Cross-sectional partial pooling uses:

```
alpha_hat_i | alpha_i ~ Normal(alpha_i, se_i^2)
alpha_i ~ Normal(mu, tau^2)
posterior mean = w_i alpha_hat_i + (1 - w_i) mu
w_i = tau^2 / (tau^2 + se_i^2)
```

Profile marginal maximum likelihood estimates a prior mean of -47 bps/year and a cross-sectional standard deviation of 129 bps/year. Shrinkage is stronger where the portfolio-specific HAC standard error is larger.

This is the key identification repair. The model no longer asks a free portfolio intercept and a persistent non-negative term to explain the same constant shift. Persistent benchmark-relative performance is represented by one estimand: alpha.

Rank and temporal validation

Rank intervals use 200 common-date circular block-bootstrap samples with 12-month blocks. Common sampled dates preserve dependence across portfolios. Rolling scores use 120-month training windows ending every 12 months. Persistence outputs state the exact overlap fraction. Forward validation freezes all training estimates, then evaluates the next 12 months only.

Secondary SFA diagnostic

The half-normal model decomposes residuals as $\epsilon = v - u$. The reported conditional score is $E[\exp(-u/\sigma) | \epsilon]$, with σ equal to the total fitted residual scale. This is invariant to expressing returns in decimals or percent. A 50:50 boundary likelihood-ratio mixture test is required before a half-normal rank is assigned. Truncated-normal results are reported only as sensitivity evidence.

4 Full-sample performance results

Rank	Portfolio	Posterior bps/year	95% interval	Bootstrap rank 95%
1	SMALL HiBM	117	[-8, 242]	[1.0, 10.0]
2	BIG LoBM	101	[20, 182]	[1.0, 9.1]
3	ME3 BM2	76	[-33, 185]	[1.0, 11.0]
4	ME1 BM4	55	[-64, 175]	[1.0, 15.0]
5	ME3 BM3	33	[-79, 144]	[2.0, 14.0]
21	ME4 BM5	-158	[-303, -13]	[16.0, 24.0]
22	ME2 BM1	-209	[-335, -82]	[19.0, 25.0]
23	ME5 BM4	-227	[-356, -98]	[19.0, 25.0]
24	ME1 BM2	-241	[-431, -51]	[20.0, 25.0]
25	SMALL LoBM	-320	[-526, -113]	[21.0, 25.0]

Five raw alpha tests survive 5% false-discovery control (5/25), but only one approximate posterior interval is wholly positive. The two statements answer different questions: FDR controls a family of raw-alpha tests, while posterior intervals reflect cross-sectional partial pooling.

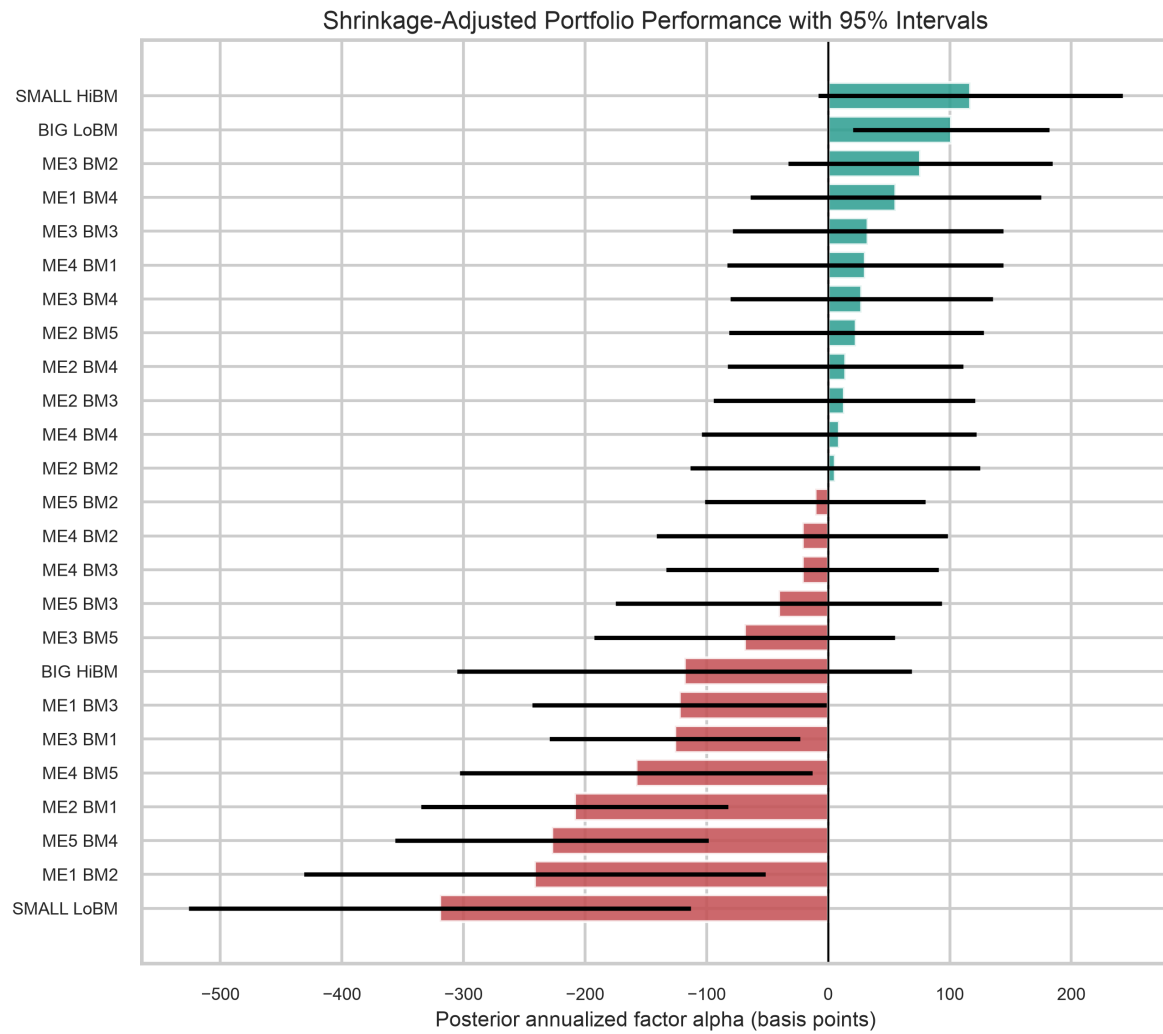


Figure 1. Posterior annualised factor alpha with approximate 95% intervals.

5 Rank uncertainty

Bootstrap uncertainty is the strongest warning against over-reading the ordered list. Even the leading portfolios can occupy a broad range of ranks when common historical blocks are resampled. The output therefore reports the probability of being in the top or bottom quintile as well as a point rank.

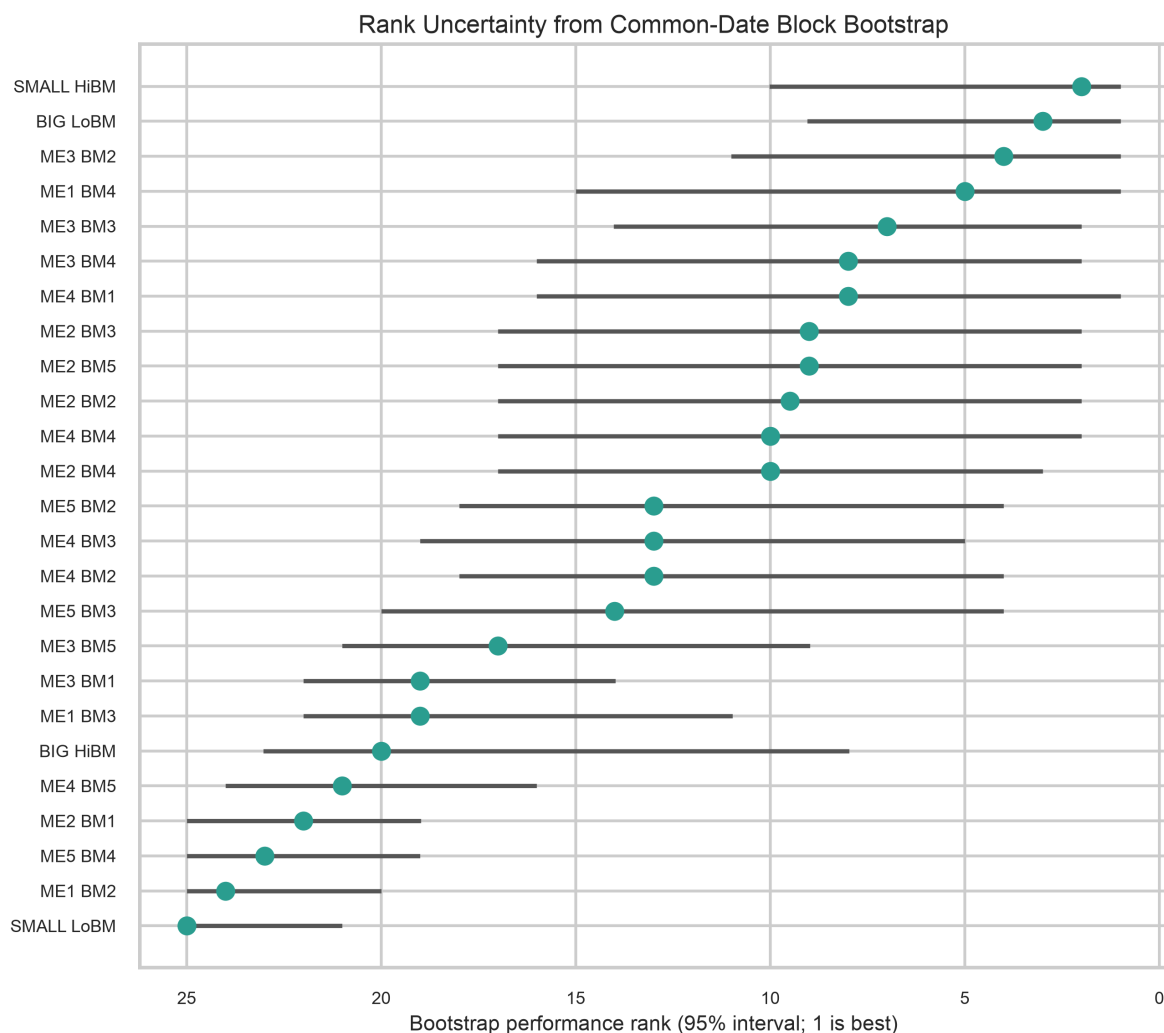


Figure 2. Common-date block-bootstrap median ranks and 95% intervals.

6 Rolling persistence without the overlap illusion

Horizon	Window overlap	Mean rank rho	Mean score r	Eligible
12 months	90%	0.877	0.894	No
60 months	50%	0.493	0.514	No
120 months	0%	0.127	0.136	Yes

The old persistence interpretation confused estimator overlap with economic stability. At a 12-month horizon, adjacent 120-month windows share 90% of their observations. At 120 months, overlap is zero and average rank persistence is only about 0.13. Only this non-overlapping comparison is labelled structurally interpretable.

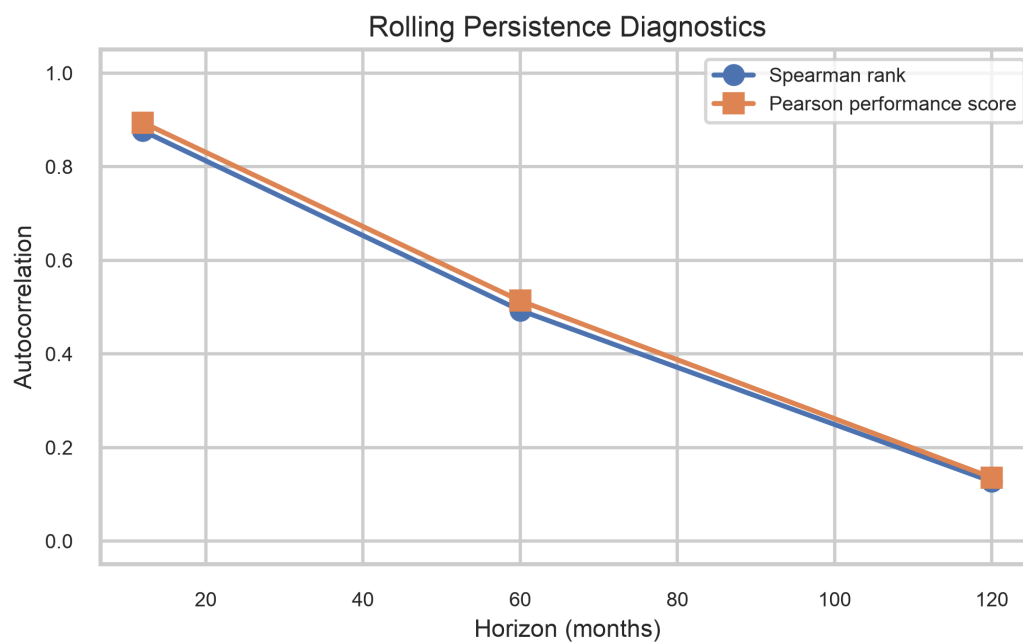


Figure 3. Rank and score persistence by calendar horizon.

7 Strictly forward validation

Metric	Estimate	HAC 95% interval	HAC p-value
Rank vs future alpha	0.096	[0.041, 0.151]	0.0006
Top - bottom future alpha	172 bps/year	[63, 280]	0.0020

There are 89 validation windows. Every evaluation month is strictly after its training window. The future spread is positive in 60.7% of windows. The result supports modest predictive ordering on average, not a uniformly successful timing rule.

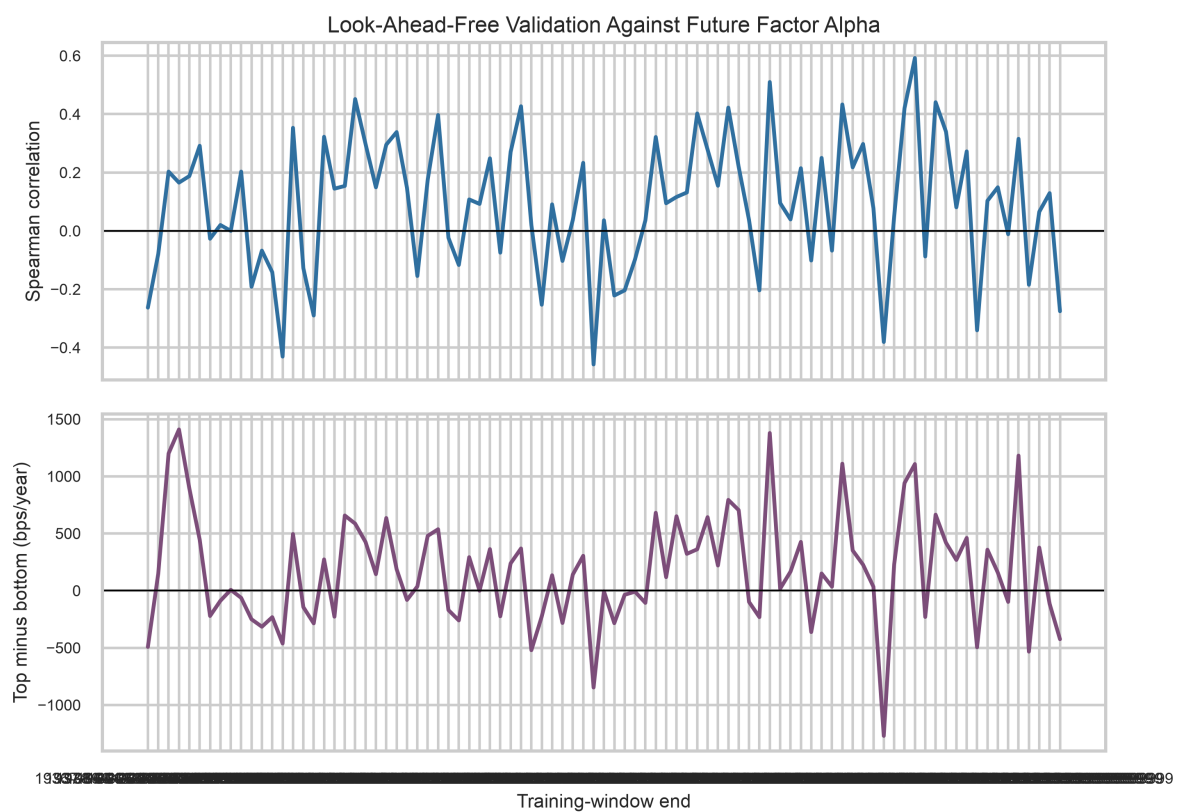


Figure 4. Window-by-window future rank correlations and top-minus-bottom spreads.

8 Model diagnostics and SFA evidence

Diagnostic	5% rejections	Interpretation
Jarque-Bera normality	25/25	Heavy tails and/or asymmetry are widespread
Ljung-Box lag 12	16/25	Serial dependence remains for many portfolios
ARCH LM lag 12	25/25	Conditional variance is time-varying

Newey-West standard errors and block bootstraps address broad dependence in inference, but they do not make the Gaussian likelihood correct. The residual tests are therefore a substantive limitation, not a footnote.

Portfolios with supported one-sided residual components

Portfolio	Boundary LR	Mixture p	Standardised AE
ME5 BM4	12.08	0.0003	0.621
ME5 BM3	6.21	0.0063	0.629

The boundary evidence is sparse. Unsupported portfolios are reported under the symmetric Gaussian null with $u = 0$ and $AE = 1$, and no SFA rank is assigned. The two supported cases indicate negative residual asymmetry under this specification; they do not prove economic inefficiency.

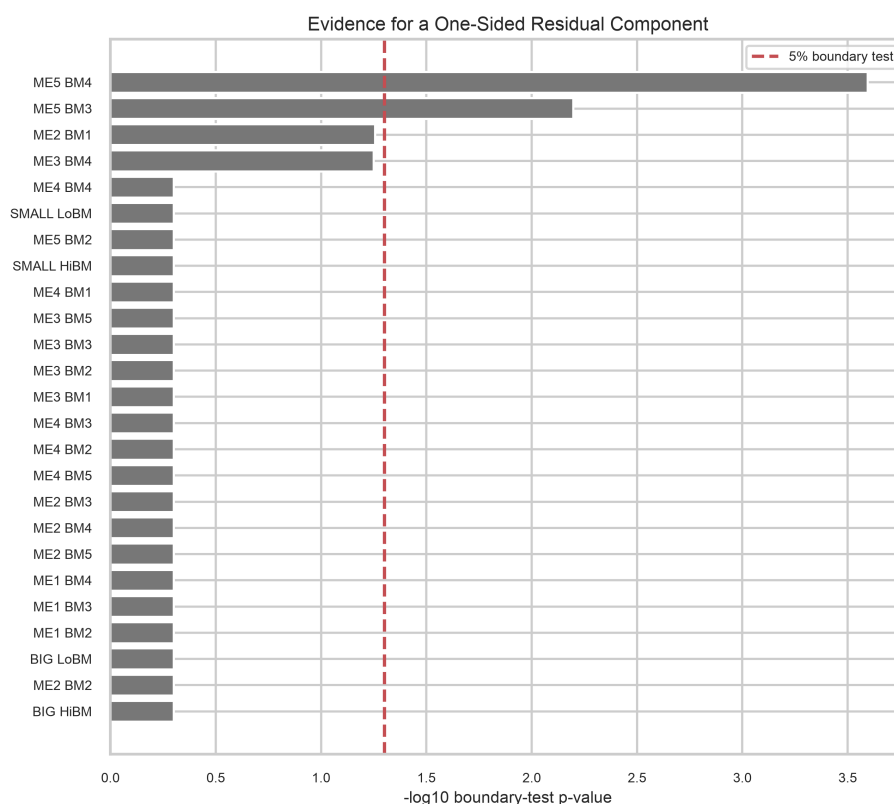


Figure 5. Boundary-test evidence for a one-sided residual component.

9 Robustness and remaining limitations

Windows compared	Rank correlation	Score correlation	Top overlap
60 vs 120 months	0.743	0.780	0.552
60 vs 180 months	0.554	0.592	0.270
120 vs 180 months	0.798	0.808	0.402

- The three-factor model is a benchmark, not a complete model. Omitted systematic risks can appear as alpha.
- Posterior intervals are empirical-Bayes approximations. Hyperparameter uncertainty is only approximated.
- The 25 constructed portfolios are cross-sectionally related. The common-date bootstrap preserves this dependence, but 200 replicates provide moderate tail precision.
- The forward evaluation periods do not overlap, but adjacent 120-month training histories do. HAC inference is used for the aggregate validation series.
- Full-sample ranks use all historical data and are descriptive. Only the forward-validation tables are out of sample.
- The SFA boundary test has limited power and depends on a restrictive distribution. The two supported cases are residual diagnostics, not managerial labels.
- Reported portfolio returns exclude implementation costs, taxes, capacity constraints, and real-time data revisions.

Overall assessment: the rebuilt project is suitable as a transparent research benchmark with explicit uncertainty. It is not a production investment signal and should not be presented as one.

10 Reproducibility and references

Reviewed execution

The codebase passes 25 tests and a full ruff lint check in a clean CPython 3.13 environment. Tests cover duplicate-key rejection, HAC recovery, shrinkage behaviour, SFA analytic gradients, a one-million-draw conditional-moment simulation, return-unit invariance, boundary suppression, overlap labelling, no-look-ahead dates, bootstrap reproducibility, and the end-to-end pipeline.

```
python -m pip install -r requirements-dev-lock.txt
pytest -q
ruff check .
MPLBACKEND=Agg python -m analysis.run_all
python -m analysis.export_tables
python reports/generate_report.py
```

Primary sources

- Fama, E. F., and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1), 3-56.
- Newey, W. K., and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703-708.
- Jondrow, J., Lovell, C. A. K., Materov, I. S., and Schmidt, P. (1982). On the estimation of technical inefficiency in the stochastic frontier production function model. *Journal of Econometrics*, 19(2-3), 233-238.
- Kenneth R. French Data Library. 25 portfolios formed on size and book-to-market and Fama-French research factors. Local review files identified by SHA-256 in data/SOURCES.md.

Canonical artefacts

Purpose	Path
Primary scores	results/tables/performance_scores.csv
Rank uncertainty	results/tables/performance_rank_uncertainty.csv
Forward aggregate	results/tables/forward_performance_aggregate.csv
Data and run manifests	results/tables/dataset_manifest.csv; results/run_manifest.json